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Author(s): M. S. Botham, R. K. Hayward, L. J. Morrell, D. P. Croft, J. R. Ward, I. Ramnarine and J. Krause

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RISK-SENSITIVE ANTIPREDATOR BEHAVIOR IN THE TRINIDADIAN GUPPY, *POECILIA RETICULATA*

M. S. BOTHAM,^{1,4} R. K. HAYWARD,¹ L. J. MORRELL,¹ D. P. CROFT,² J. R. WARD,¹ I. RAMNARINE,³ AND J. KRAUSE¹

¹*Institute of Integrative and Comparative Biology, University of Leeds, Clarendon Way, Leeds, West Yorkshire LS2 9JT United Kingdom*

²*School of Biological Sciences, University of Wales Bangor, Bangor, Gwynedd LL57 2UW United Kingdom*

³*Department of Life Sciences, University of the West Indies, St. Augustine, Trinidad and Tobago*

Abstract. The comparative approach has become a powerful tool for understanding how predation has shaped prey behavior. In this study we recorded the occurrence of common aquatic predator species and their densities in seven natural populations of Trinidadian guppies, *Poecilia reticulata*. We then exposed shoals of guppies from each of these populations to a series of predator treatments. Predator treatments differed in the species of predator used (pike cichlids, *Crenicichla frenata*; rivulus, *Rivulus hartii*; and freshwater prawns, *Macrobrachium carcinus*) and thus in the level of risk posed. We recorded the antipredator responses of guppies in each of these predator treatments.

The strength of antipredator behavior shown by guppies was affected by both the type of predator they were exposed to and the level of predation risk they experienced naturally in the wild. Importantly, we found that guppies from high-risk populations showed a heightened response, compared to those from lower risk populations, only when exposed to the predator species that posed the greatest risk. Our results show the importance of individual predator species in shaping the behavioral traits of prey species at the population level. This has implications for prey movement between habitats that are geographically close but differ greatly in predator fauna.

Key words: antipredator behavior; *Crenicichla*; guppy; multiple predators; *Poecilia reticulata*; predator recognition; *Rivulus*.

INTRODUCTION

Most animals are faced with a wide range of predators that hunt using various strategies, which themselves may vary spatially and/or temporally. To increase their chances of survival in multi-predator environments, prey often exhibit a range of both general and specific antipredator responses. Antipredator behaviors are associated with both direct energetic costs, which depend upon the duration and intensity of the behavior, and indirect costs to fitness incurred through the loss of time devoted to other activities such as foraging and mating (Lima and Bednekoff 1999). For this reason, prey not only exhibit the appropriate behavior for a given type of predator, but also moderate the duration and/or intensity of these behaviors depending on the perceived level of threat posed by the predator(s) present (known as risk-sensitive behavior; Lima and Bednekoff 1999).

To ascertain risk, prey must discriminate between different predator types and/or the motivational states of predators they are faced with. For many animals, this may require experience with each of the different predators either directly through an interaction with

them (Griffin et al. 2001) or indirectly through mechanisms such as social learning (Brown and Laland 2003). Alternatively, experience with any type of predator may be sufficient for the development of general antipredator responses that a prey may exhibit toward all predators regardless of whether or not it has had experience with them (Blumstein 2006). This is known as the multi-predator hypothesis (Blumstein 2006), and may explain why many prey are able to respond to predators that they are not sympatric with. Under these circumstances, prey from populations exposed to high predation risk, regardless of the types of predators, are likely to show stronger antipredator responses than those from populations with a low-predation risk (e.g., Seghers and Magurran 1995).

Predators exert evolutionary pressure on a prey's antipredator behavior, and so prey populations often differ in how well developed their antipredator responses are, even in the absence of a predator (Magurran and Seghers 1990a). However, predators differ, between species, in the level of risk they pose to a given prey (Lima 1992). Prey able to distinguish between these different predators should alter their antipredator responses in a risk-sensitive manner. Population differences in antipredator behavior, therefore, are likely to be enhanced or reduced depending on whether the prey is familiar with the predator or not (e.g., Kelley and Magurran 2003, Murray et al. 2004). To fully under-

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⁴ Present address: Centre for Ecology and Hydrology, Monks Wood, Abbots Ripton, Huntingdon, Cambridgeshire PE28 2LS United Kingdom. E-mail: bgymsb@gmail.co.uk

TABLE 1. Details of the locations of the seven rivers in which we assessed predation risk, with their associated predation ranks (calculated in 2006 and 2005) and predator densities (number of individuals within four predator body lengths of the prey stimulus per minute).

River	Grid reference		Predation rank		Predator density							
					Pike cichlids		Acaras		Rivulus		Freshwater prawns	
	North	West	2005†	2006	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Aripo	10°40'	61°14'	1	1	2.27	0.49	0.02	0.01	0.00	0.00	0.00	0.00
Tacarigua	10°41'	61°22'	2	2	1.31	0.57	0.24	0.06	0.00	0.00	0.00	0.00
Arima	10°41'	61°17'	4	3	0.46	0.28	0.22	0.22	0.00	0.00	0.00	0.00
Lower Turure	10°40'	61°10'	3	4	0.35	0.18	0.00	0.00	0.00	0.00	0.00	0.00
Arouca	10°40'	61°19'	5	5	0.07	0.07	0.00	0.00	0.00	0.00	0.00	0.00
Naranjo	10°41'	61°14'	7	6	0.00	0.00	0.00	0.00	1.23	0.53	0.00	0.00
Upper Turure	10°41'	61°10'	6	7	0.00	0.00	0.00	0.00	0.64	0.33	0.06	0.04

† See Croft et al. (2006).

stand antipredator behavior, therefore, the interaction between the natural predator regime of a prey population and the predator with which a prey is faced at a given point in time should be considered. The comparative approach, whereby multiple populations of the same or closely related prey species, which differ in the predation risk they experience, are compared with one another, makes this possible (e.g., Magurran and Seghers 1990a, Mathis et al. 1993). Studies using this approach have investigated differences in the level of antipredator behavior as a function of predation risk between populations classed as high and low/no predation (e.g., Owings et al. 2001, Magurran 2005, Magnhagen 2006). A more quantitative assessment of predation risk in which both the composition and density of the natural predator regime is required to fully investigate the interaction between population predation risk and predator-specific antipredator behavior. To our knowledge, no detailed study has yet attempted this.

The Trinidadian guppy, *Poecilia reticulata*, is an excellent model organism to investigate the evolutionary role of predation due to its widespread abundance in a diverse range of freshwater habitats and communities (Magurran 2005). In the northern mountain range of Trinidad, separate populations of guppies experience different types and densities of aquatic predators (Magurran 2005), as natural barriers such as waterfalls prevent the upstream migration of predators. As a result, it has been possible to make comparisons of the behavior (e.g., Magurran and Seghers 1990a), morphology (e.g., Odell et al. 2003), and life history (Reznick and Endler 1982) of guppies as a function of the predation level they experience. Traditionally, guppy populations have been classed as either "low" or "high" depending on the presence or absence of a particular predator, the pike cichlid, *Crenicichla* sp. (Magurran 2005), yet it is known that both the number and type of predators can greatly influence the way in which prey behave (Relyea 2003, Vance-Chalcraft et al. 2004), and predator assemblages differ greatly between sites within the same risk category (Botham 2005). Recently, Croft et al. (2006) proposed a methodology for ranking multiple

rivers on a continuum of predation risk, based on the abundance of certain predator species, rather than classifying rivers as high or low. Here, we utilized this method in order to investigate the behavioral responses of guppies from multiple rivers to different predators, in relation to both the risk posed by that specific predator, and the level of risk they have naturally experienced.

Here we investigated the following hypotheses: (1) antipredator responses (such as shoaling and vigilance) will be stronger in the presence than in the absence of a predator in all populations; (2) the strength of the antipredator responses will depend on the predator species present; (3) prey responses will depend on the interaction between the population's history of natural predation risk (level and/or type of risk) and the predator species present.

MATERIALS AND METHODS

To investigate the effect of immediate risk and prior experience on the antipredator behavior of guppies, we used both a field and a laboratory study. Guppies show strong sexual dimorphism both morphologically and behaviorally, leading to male-biased predation (Magurran 2005). Females show stronger shoaling tendencies and are more wary of predators than males (Magurran et al. 1992). We therefore used only female guppies, as this is likely to provide a more conservative estimate of predation risk in the field (Croft et al. 2006), eliminates any confounding effects of sexual behavior, and is likely to generate stronger and more risk-sensitive antipredator responses (Magurran and Nowak 1991).

Investigating predation risk in the field

Study sites.—Seven rivers (Table 1) from the northern mountain range of Trinidad were sampled during May and June 2006, at the end of the dry season, to determine the predation risk associated with different rivers. We followed the approach of Croft et al. (2006) and assessed predation risk by ranking the rivers on a continuum. This method is a relatively simple way to quantitatively assess predation risk that shows consistency with previous qualitative measurements (Seghers 1973, Croft et al. 2006). The seven rivers we selected were a subset of those

previously studied (Croft et al. 2006) to allow us to assess the consistency of predation risk over time. Pike cichlids (*Crenicichla* sp.), regarded as the most dangerous guppy predator, have been recorded in five of the seven rivers. These rivers (the Lower Aripo, Tacarigua, Lower Ture, Arima, and Arouca) are traditionally classed as high predation (Magurran 2005). Other predators, such as blue acara cichlids (*Aequidens pulcher*) and wolf fish (*Hoplias malabaricus*) are also found in these rivers. The remaining two rivers (Naranjo and Upper Ture) are traditionally classed as low predation, as they do not contain these predatory species. Hart's rivulus (*Rivulus hartii*) is present in most rivers, but is the only potential predatory fish at low-predation sites (Magurran 2005), and is likely to pose a danger only to smaller guppies (Mattingly and Butler 1994). A further potential predator, the freshwater prawn (*Macrobrachium* spp.) is also found in these low-predation sites, but the extent of the risk these pose to guppies is unknown (Magurran 2005). Guppies from "low-predation" rivers have been shown to display weaker antipredator responses, such as shoaling and predator inspection behavior, than those from rivers containing larger predators such as pike cichlids (e.g., Magurran and Seghers 1990a, Seghers and Magurran 1995).

Assessing predation risk.—To establish the relative predation risk in the seven rivers, we recorded the species and frequency of predators approaching and orientating toward a stimulus female guppy, presented in a transparent, cylindrical container.

Stimulus females (body length 25–30 mm) were caught using a small beach seine net (2 m long, 1 m high, 3 mm mesh size) from a different pool in the same river in which they were presented. A single female was placed in a transparent, cylindrical container (65 mm diameter, 130 mm high), pierced with over 50 holes of ~1 mm diameter, allowing the transmission of both visual and olfactory cues. The cylinder was weighted with substrate collected from a nearby pool from the same river in which the assessment took place, so that it remained stationary on the riverbed throughout observations, and was suspended from monofilament line to allow easy positioning and removal from the water. Each stimulus guppy was used in only one pool, following which it was returned immediately to the pool from which it was caught.

We placed a stimulus guppy in a total of five pools in each of the seven rivers studied. We selected pools that contained a reasonable level of habitat heterogeneity, including rocky refuges and shallow areas with peripheral vegetation, which could be used as refuges by predators (Botham 2005). Pools also contained a deep area (>30 cm), as predators are known to be more abundant in deeper water (Botham 2005), which could be easily observed from the river bank. The cylinder containing the female guppy was placed slowly into the deepest part of the pool using a rod (~2 m long) to position the container.

Following the positioning of the stimulus guppy in the river, we allowed a period of 10 min for any predators in the pool to acclimate to the container and the presence of an observer on the bank, and for any disturbed sediment to settle. The observer was positioned on the river bank, at least 2 m from the stimulus container, and relayed observations to a second person positioned out of sight of the container. After the acclimatization period, we recorded the number and identity of predators that approached the stimulus guppy (within four predator body lengths) every 10 s for 10 min, giving a total of 60 point samples for each pool for each river.

These presentations were made at midday under full daylight conditions, and as such, are likely to provide a very conservative measure of how many freshwater prawns were present given their more nocturnal behavior (Magurran 2005). However, our primary aim was to ascertain the relative predation risk guppies are faced with by each predator species during daylight hours. While freshwater prawns were present and active in a number of rivers throughout our presentations, they rarely confronted the stimulus guppy, suggesting they may not be an important guppy predator (see Magurran 2005).

Ranking rivers on a continuum of risk.—We observed and recorded the presence of pike cichlids (*C. frenata*), acara cichlids, *Aequidens pulcher*, Hart's rivulus (*R. hartii*), and freshwater prawns (*Macrobrachium* spp.) around the stimulus female. Although we observed other potential predators, including wolf fish (*Hoplias malabaricus*), catfish (*Rhamdia quelen*), and two species of large characins (*Astyanax bimaculatus* and *Hemibrycon* sp.), these never approached the stimulus female, potentially as a result of spatial and temporal differences in habitat use (Seghers 1973). Following Croft et al. (2006), we ranked the rivers using a system based on the presence and abundance of the different predators. Rivers were first assessed according to the presence of the most dangerous predator, pike cichlids (*C. frenata*; Kelley and Magurran 2003, Botham et al. 2006). Where this predator was present, rivers were ranked in decreasing order of predator abundance, such that the river ranked at 1 contained the highest abundance of pike cichlids. Once all rivers containing pike cichlids had been ranked, we looked for the abundance of acara cichlids (*A. pulcher*), followed by Hart's rivulus (*R. hartii*), and finally freshwater prawns, (*Macrobrachium* spp.). While this method is weighted toward the presence of pike cichlids, all the rivers studied that contained acaras also contained pike cichlids. In addition, acaras are omnivores that only occasionally attack guppies, whereas pike cichlids are carnivorous and are major guppy predators (Botham 2005).

Risk-sensitive responses in the laboratory

The aim of the laboratory study was to investigate how guppies from different rivers, subject to various levels of predation risk in the wild, behaved in response

to three common aquatic predators that differed in the risk they posed. We caught 200 female guppies (25–35 mm total length; TL) from each of the seven rivers in which we assessed the predation risk in the field. Guppies were caught using a small beach seine net (2 × 1 m, 3-mm mesh size) from pools in which we made predator observations and from adjoining pools with similar habitat and structure. Each river population was kept in a separate pool (160 cm diameter), covered with a fine mesh, and fed ad libitum on dried flake food.

Predator species and holding conditions.—As predators, we chose three species that vary in the level of risk they pose to guppies and that have different hunting strategies. Previous work has shown that guppies show risk-sensitive responses to predatory fish (Botham et al. 2006). Because predators within the same taxon may elicit similar types of antipredator response, we chose to investigate the responses of guppies both to different predators from within the same taxon that differ in the risk they posed, and to a predator from a different taxonomic group. We used pike cichlids, *C. frenata*, as the high-risk fish predator and rivulus, *R. hartii*, as the low-risk fish predator (Magurran 2005). As a third predator species, we used freshwater prawns, *M. carcinus*. Predatory fish and invertebrate predators often elicit different antipredator responses in prey (e.g., Soluk and Collins 1988, Turner et al. 1999).

We caught five adult pike cichlids (170–190 mm TL) from the Tacarigua River using rod and line with small barb-less hooks baited with dead female guppies. Female guppies were caught using a small beach seine net (2 × 1 m, 3-mm mesh size) and were killed with a single tap to the head. Six rivulus (110–120 mm TL) were caught from the Paria and Upper Turure rivers using a small hand seine net (2 × 1 m, 3-mm mesh size), and three freshwater prawns (110–120 mm TL) from the Paria River at dusk using dip nets (30 × 20 cm, 1-mm mesh size). All predators were kept in separate aquaria (500 × 300 × 300 mm, filled to a depth of 200 mm). Each aquarium contained a gravel substrate (depth 15 mm), a small rock refuge, and a floating polystyrene refuge. Predators were fed daily on female guppies (e.g., Brown and Dreier 2002). Aquaria were aerated with small corner filters. Both predators and prey were maintained under a 12-h light : 12-h dark regime at 25°C to replicate conditions in the wild.

Experimental set-up and procedure.—We set up four experimental pools (160 cm diameter), one for each predator treatment. The four treatments were as follows: (1) pike cichlid, (2) rivulus, (3) freshwater prawn, and (4) no predator (control). The treatment used in each pool was always the same to avoid mixing of odor cues from the predators. In each experimental pool, we marked a circle 20 cm from the edge of the pool using a permanent marker pen to mark the zone in which shoals of test guppies were released at the start of experimental trials. All pools were filled with aged water to a depth of 10 cm. Two hides were positioned on opposite sides of the pool,

to conceal the observers. The pool was lit using four 40-W lamps located in front of the two hides. In the center of the pool, we positioned a predator container. This was a transparent cylinder (20 cm diameter × 50 cm height), perforated to allow the transmission of both visual and olfactory cues from the predator into the experimental pool.

At the beginning of each experimental trial, an individual of the appropriate predator species was captured from its holding tank using small hand nets, and placed into the central predator container. In addition, we poured 500 mL of water from the predator's holding tank into the container. For the control treatment, no predator was placed in the central predator container and 500 mL of aged water was poured in. After the predator had been placed in the pool, we caught a shoal of 10 female guppies, approximately size assorted, from one of the stock pools, using small dip nets. The shoal was placed in an opaque cylinder (65 mm diameter, 130 mm height) positioned in the release zone in the experimental pool at the start of each experimental trial. Five shoals of guppies from each of the seven river populations were tested under each of the four different predator treatments (a total of 20 shoals per population). Each shoal of stimulus fish was only used once, and thus was subject to only one predator treatment. Following the completion of a trial, each shoal was placed into a population stock tank set up for fish that had been tested.

After 3 min, the opaque cylinder containing the stimulus fish was removed using a remote pulley system to release the shoal into the experimental pool while the observers remained behind their hides. Following their release, we allowed 10 min for the guppies to acclimate to the experimental pool and to recover from any disturbance caused. In the last minute of this acclimatization period, a third person, blind to the treatment under which the guppies were being tested, randomly selected a focal individual for observer 1 to make behavioral observations on. Following the acclimatization period, we recorded the behavior of the focal individual every 20 s for 10 min. Observer 1 relayed information to observer 2 for recording, allowing observer 1 to maintain continuous visual contact with the test fish. We recorded whether the focal individual was engaged in one or more of the following four behaviors: (1) Shoaling. The focal individual was considered to be shoaling if it was within four body lengths of a conspecific (Pitcher and Parrish 1993). Guppies are thought to increase their shoaling behavior in response to perceived predation threat (e.g., Botham et al. 2006). (2) Foraging. The focal individual was considered to be foraging if it made pecking movements toward a potential food item. Although the fish were not intentionally fed during the trial, guppies were observed to forage on algae growing within the pool, as well as small insects that had fallen into the pools. Guppies are

known to reduce their foraging behavior when threatened by a predator (Dugatkin and Godin 1992, Fraser et al. 2004). (3) Position in water column. We recorded whether the focal individual was positioned at the surface of the water (upper third), on the bottom (lower third), or in the middle of the water column (middle third). As predators are most common in deeper water, increased time at the surface may represent an antipredator behavior (Botham et al. 2006). (4) Orientation. Orientation toward a predator represents vigilance and may represent the first stage, or a basic form, of predator inspection behavior, and suggests that the prey fish may be wary of the predator and recognize it as a threat (Kelley and Magurran 2003).

Statistical analysis

Statistical analysis was carried out in SPSS version 11.5.0 (SPSS 2002) and R version 2.3.1 (R Development Core Team 2006). Firstly, we carried out Spearman rank correlations between response variables in the control treatment to check for independence. There were no significant correlations between any of the response variables; therefore, we treated each variable independently in subsequent analyses. For the main analysis, investigating the effect of predation risk (river rank), predator treatment (control, pike cichlid, freshwater prawn, and rivulus), and the interaction between these variables, on the behavior of the focal individual, we used generalized linear modeling. The river variable took the values 1 to 7, ordered by decreased predation risk, and was treated as a linear covariate (see Croft et al. 2006). The proportion of time spent shoaling, at the surface, feeding, and orientated toward the predator, by the focal individual, were treated as independent response variables, with quasibinomially distributed error variances (to account for overdispersion in the data) and logistic link functions.

We found no significant interactions between natural predation risk and any predator treatment for the proportion of time spent foraging, and time spent orientated toward the predator. Consequently, these interactions were dropped from the analysis to produce the minimum adequate model. Significant interactions were found in the remaining analyses, and these are presented in full. The results are presented in Table 2.

As the tests compare the effect of predator treatments to the control, we needed to look for differences between predator treatments (e.g., Is the response to predator A greater than the response to predator B?) to test our a priori prediction that the type of predator present would affect the strength of antipredator response. Following Crawley (2005), we first looked for the difference between the mean values for the two treatments under consideration ("Estimate" in Table 2). We then compared this difference to the sum of the standard errors ("Error" in Table 2) of the same treatments. If the difference between the estimates for two treatments is greater than the sum of their standard errors, this

suggests a significant difference between the treatments at the $P < 0.05$ level (Crawley 2005). Using this information, we determined which treatments were significantly different from one another for each behavioral response measured and then rearranged the order in which treatments were compared in our original generalized linear models to acquire exact P values for these differences. Details of statistically significant differences between predator treatments are presented in the results section. The results for all comparisons between predator treatments (except those omitted following a significant interaction term) can be found in the Appendix.

RESULTS

Investigating predation risk in the field

Table 1 shows the order in which we ranked the seven rivers based on the abundances of pike cichlids, acaras, rivulus, and freshwater prawns observed within four predator lengths of a stimulus guppy in each river. Included in this table is the ranking assigned to the same seven rivers by Croft et al. (2006) calculated a year earlier in 2005 using the same technique. Predation risk in each of the rivers was significantly correlated over the two years (Spearman rank correlation: $r_s = 0.929$, $P = 0.003$, $N = 7$). These results suggest that our technique to assess predation risk is robust, and that differences in predation risk between rivers persist over time, allowing us to relate the variation in antipredator behavior between guppy populations to the differences in the level of natural predation risk they are subject to.

Risk-sensitive responses in the laboratory

Predator behavior.—We observed no difference in the activity levels of the three different predator species used in this study. All predators held station under the floating refuge provided in their containers during the experimental trials, and made no attempts to attack guppies or to escape their containers.

Prey responses to different predators

Foraging behavior.—The cross-population comparison revealed differences in the time guppies spent foraging between rivers that is associated with the respective differences in predation risk, "population predation risk," (Table 2, Fig. 1a): As predation risk decreased, the time spent foraging by guppies increased. Guppies spent significantly less time foraging in the pike cichlid and rivulus treatments than in the control (Table 2, Fig. 1a), but there was no significant difference in foraging time between the control and freshwater prawn treatments (Table 2).

Shoaling behavior.—Unlike with foraging behavior, there was no effect of population predation risk between the different populations on the time spent shoaling (Table 2, Fig. 1b). However, we found a significant interaction between population predation risk and predator treatment (Table 2, Fig. 1b). This was due to

TABLE 2. Results of the generalized linear modeling analyses of the effects of predation risk and predator treatment on four independent response variables.

Response variable, risk, and treatment	Estimate	Error	<i>t</i>	<i>P</i>
Time spent foraging				
Risk	0.215	0.075	2.879	0.005**
Treatment				
Pike cichlid	−1.291	0.436	−2.963	0.004**
Prawn	−0.369	0.327	−1.132	0.260
Rivulus	−1.861	0.544	−3.420	<0.001***
Time spent shoaling				
Risk	0.018	0.052	0.346	0.730
Treatment				
Pike cichlid	4.128	0.790	5.227	<0.001***
Prawn	0.848	0.372	2.278	0.024*
Rivulus	1.136	0.402	2.823	0.005**
Interaction				
Risk × pike cichlid	−0.420	0.145	−2.895	0.004**
Risk × prawn	−0.015	0.084	−0.175	0.861
Risk × rivulus	0.007	0.091	0.077	0.939
Time spent near surface				
Risk	0.018	0.111	0.160	0.873
Treatment				
Pike cichlid	3.250	0.622	5.223	<0.001***
Prawn	0.186	0.687	0.267	0.790
Rivulus	1.820	0.606	3.002	0.003**
Interaction				
Risk × pike cichlid	−0.340	0.138	−2.453	0.015*
Risk × prawn	0.005	0.152	0.036	0.971
Risk × rivulus	−0.031	0.135	−0.232	0.817
Time spent oriented toward predator				
Risk	−0.092	0.059	−1.578	0.117
Treatment				
Pike cichlid	2.227	0.442	5.037	<0.001***
Prawn	1.391	0.463	3.006	0.003**
Rivulus	1.318	0.466	2.829	0.005**

Notes: Interaction terms relate to the interaction between predation risk associated with each guppy population and the predator treatment under which they were exposed ($df = 132$). Where no significant interactions were detected, these were removed from the analysis, and main effects only are presented ($df = 135$). Significant effects are highlighted in boldface, and significance levels are illustrated as follows: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

the pike cichlid treatment: As predation risk in rivers decreased, the time spent shoaling by guppies also decreased in the presence of a pike cichlid. Guppies showed an increase in shoaling behavior when presented with either a rivulus or a freshwater prawn compared to the control (Table 2, Fig. 1b).

Position in the water column.—We found that there was no overall effect of population differences in predation risk on how long guppies spent in the surface water (Table 2). As with time spent shoaling, we found a significant interaction between population predation risk and predator treatment, due to the pike cichlid treatment, for the time which guppies spent in the surface water (Table 2, Fig. 1c). As the population predation risk decreased, guppies reduced the time they spent near the surface when faced with a pike cichlid. When faced with a rivulus, guppies spent significantly more time near the surface than in the control treatment, but not when faced with a freshwater prawn (Table 2,

Fig. 1c). In addition, guppies spent significantly more time in the surface water when faced with a rivulus than when faced with a freshwater prawn (GLM, $df = 132$, $t = -2.832$, $P = 0.005$; see Fig. 1c).

Orientation toward the predator.—Again, there was no overall effect of population predation risk on the time spent orientated toward the predator container by guppies (Table 2). Guppies spent significantly more time orientated toward the predator container in all predator treatments compared to the control (Table 2, Fig. 1d). Although the time spent orientated toward the predator was greatest in the pike cichlid treatment (see Fig. 1d), there was no significant difference in orientation behavior between any of the predator treatments.

DISCUSSION

We found that risk-sensitive antipredator behavior is affected by an interaction between the type of predator a prey is exposed to, and by the level and type of

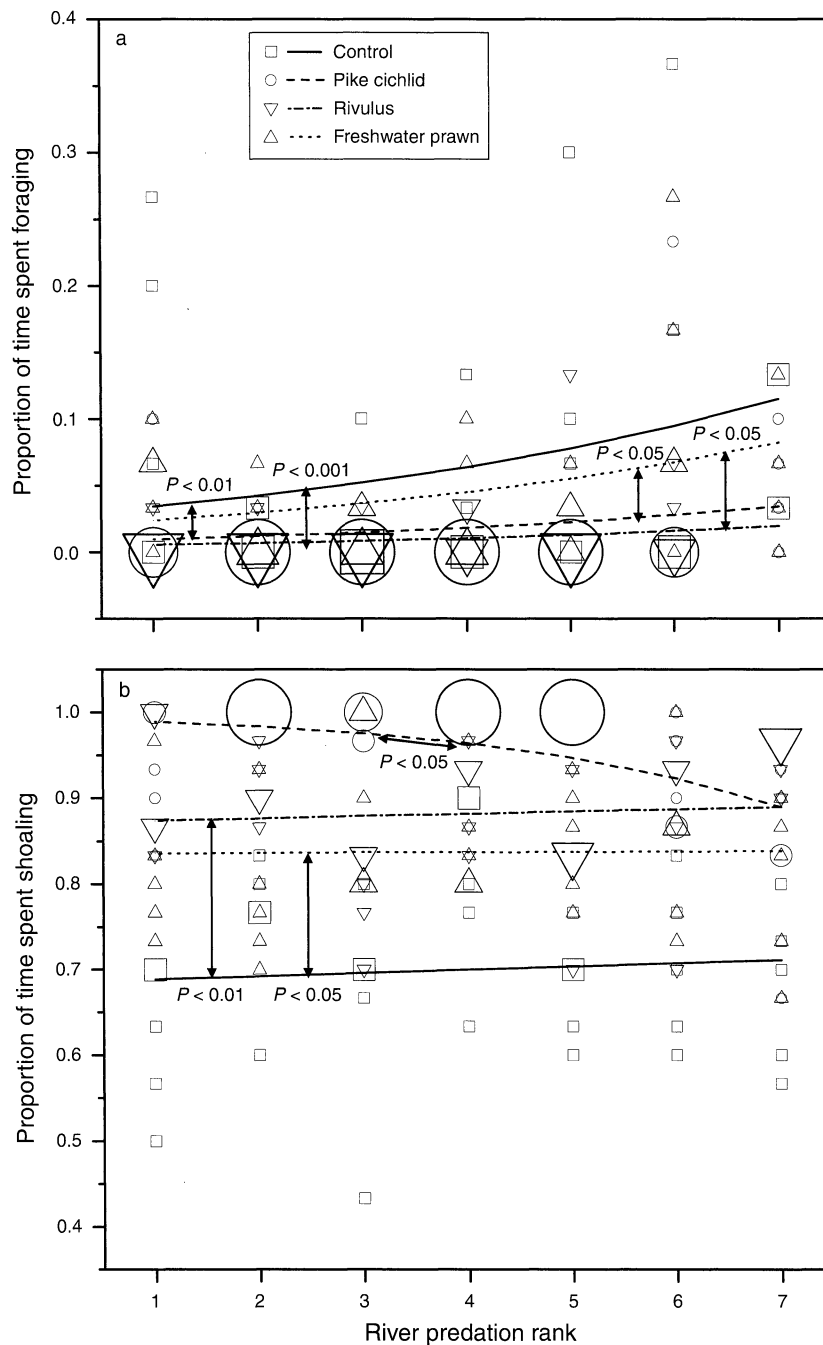


FIG. 1. The proportion of time guppies from each of seven different river populations spent (a) foraging, (b) shoaling, (c) at the surface, and (d) orientated toward the predator container, under four predator treatments (control, pike cichlid, rivulus, and freshwater prawn). Predation rank is labeled on the x-axis starting with the highest ranked river (Aripo) at the left through to the lowest ranked river (Upper Ture) on the right. Horizontal arrows indicate significant interaction terms, and vertical arrows indicate significant differences between predator treatments, with their respective P values. The size of each point corresponds to the number of points with the same value.

predation risk it experiences in the wild. Furthermore, we found evidence that antipredator responses in guppies may differ with predatory taxa; some of the typical responses to predatory fish (e.g., Magurran 2005, Botham et al. 2006) were not performed in the presence

of freshwater prawns. Finally, a comparison with previous assessment of predation risk in the wild (Croft et al. 2006) showed that the number and species assemblage of predators in different populations is consistent over time.

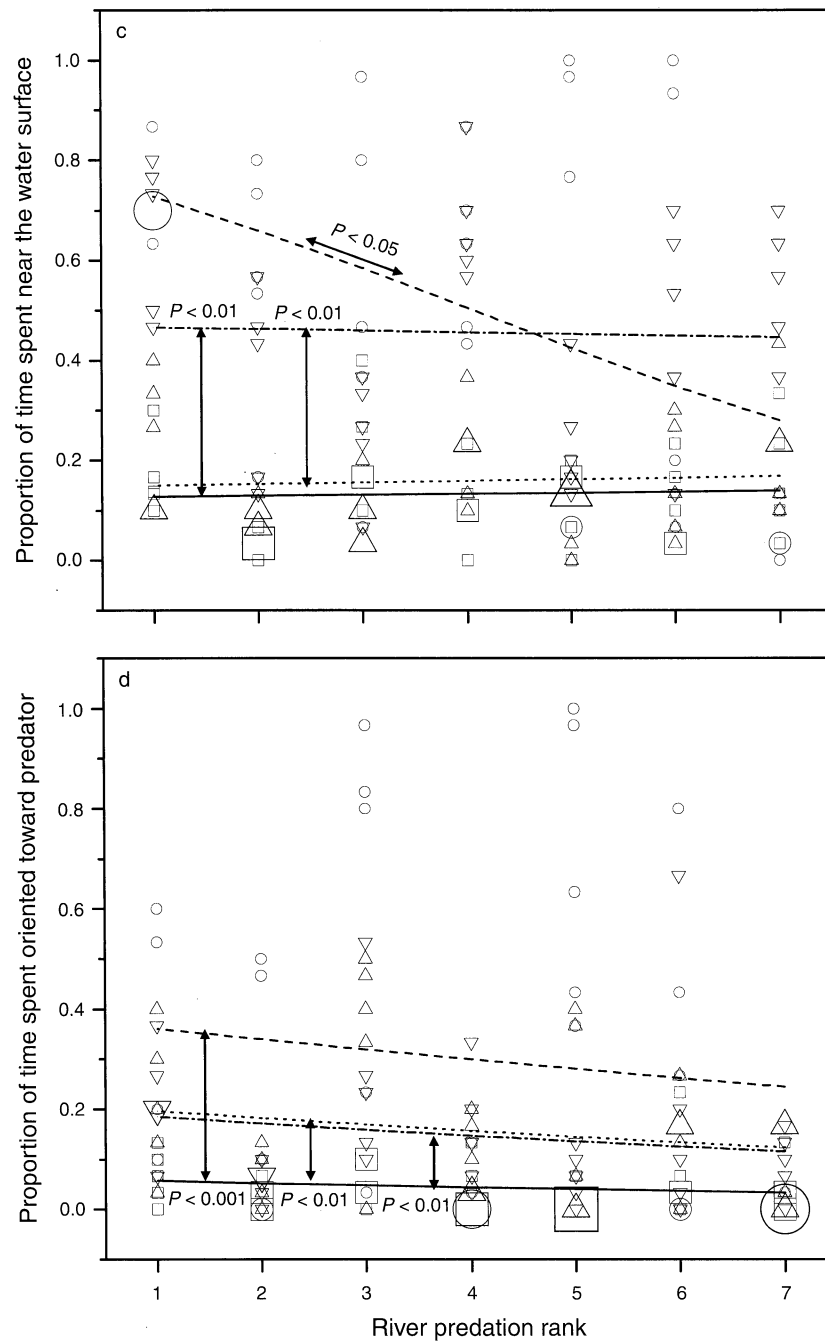


FIG. 1. Continued.

*The effect of natural predation risk
on antipredator behavior*

Many studies have shown that there can be marked differences in the antipredator responses of animal populations that experience different levels of predation even in the absence of a predator (e.g., Owings et al. 2001 [mammals], Brown 2002 [fish], Husak and Rouse 2006 [lizards]). Such differences have been shown to

exist between guppy populations (e.g., Magurran and Seghers 1990a, Seghers and Magurran 1995). In the current study, only time spent foraging was significantly affected by natural predation risk in the absence of a predator. This difference may be an adaptive response in the guppies' temporal pattern of foraging to high densities of diurnal predators. In high-predation rivers, guppies may spend less time foraging in the day to

remain vigilant to predators and compensate by foraging throughout the night (Fraser et al. 2004). The other behaviors measured did not significantly differ as a function of natural predation risk alone. In addition, Fig. 1 shows a linear relationship between the time allocated to foraging and natural predation risk, suggesting the variation between the populations in foraging behavior is not primarily caused by differences in predation risk, but is likely to be a result of other factors such as social ones (Blumstein et al. 1999). Conversely, it may more simply be that not all behaviors were sensitive to natural predation risk. When assessing group size effects in tamar wallabies, *Macropus eugenii*, Blumstein et al. (1999) found that perceived predation risk was primarily responsible for the variation in time allocated to most, but not all, of a range of behaviors.

However, all behaviors were significantly affected by the presence of predators, suggesting that they were sensitive to immediate predation risk. When faced with a fish predator, guppies showed a decrease in foraging activity, and an increase in the time spent shoaling, in the surface water and orientated toward the predator. Time spent shoaling and orientated toward the predator was also significantly greater in the freshwater prawn treatment than in the control. Most importantly, we found that for shoaling behavior and time spent in the surface water, there was a significant interaction between the pike cichlid treatment and natural predation risk, whereby guppies responded much more strongly to pike cichlids if they were from populations that contained pike cichlids compared to guppies from lower risk rivers that did not contain pike cichlids. There was no difference in behavioral responses between rivulus and pike cichlid treatments for guppy populations lacking pike cichlids in their natal rivers, whereas guppies from populations sympatric with pike cichlids spent more time shoaling and in the surface waters when faced with a pike cichlid than when faced with a rivulus. This suggests that, while all populations are capable of responding to potential predators per se, encounters with different predator species are required to hone predator-specific responses and behave risk sensitively toward different predator species within the same taxa. This interaction has important implications for the range expansion of prey species, as it demonstrates how prey require experience with a complete range of predators to respond to them in a species-specific manner. While the presence of other less dangerous predators may be enough to enable prey to respond to predators with which they have had no experience (Blumstein 2006), if they encounter new and more dangerous predator species, they may fail to respond strongly enough to avoid being killed.

The effect of different predators on antipredator behavior

Differences in hunting strategy between predator species often result in prey adopting antipredator

responses that are specific to each predator type (Sih et al. 1998). This occurs across all taxa (for example; mayfly larvae vs. fish and stonefly larvae [Soluk and Collins 1988]; fish vs. predatory birds and fish [Templeton and Shriner 2004]; and primates and lizards vs. raptors and snakes [Seyfarth et al. 1980, Stapley 2004]). In addition, because different predators exert different predation pressures, it is beneficial for prey animals to adjust the duration, intensity, and type of response depending on the level of threat (e.g., Puttlitz et al. 1999, Banks 2001, Templeton et al. 2005). Therefore, risk-sensitive behavior requires animals to assess the level (how dangerous a predator is) and type (different predator species) of risk. Key to this assessment is the recognition of different predators and the risk they each pose.

Animals often use simple visual cues such as the shape, size, and movement of another animal (Engstrom-Ost and Lehtiniemi 2004) and olfactory cues associated with its diet (Wisenden 2000) to determine whether it is a predatory threat. Predators were fed on equal numbers of guppies prior to experimental trials, therefore, olfactory cues associated with diet are unlikely to have differed between predator species. Similarly, we noticed no differences in activity between the different predators used in this study. Shape and size of the predators used in this study did differ between different predator treatments, especially between the freshwater prawn and two fish predator treatments. The size of an animal and/or the size of its mouth are simple and reliable visual cues to assess the risk of potential predators (Engstrom-Ost and Lehtiniemi 2004). For example, reef fish use the distance between the eyes and the size of the mouth to discriminate between piscivorous and non-piscivorous fish (Karplus and Algom 1981), and Black-capped Chickadees, *Poecile atricapilla*, increase the intensity of mobbing behavior toward larger raptor species (Templeton et al. 2005). While the time allocated to antipredator behaviors was greatest in the pike cichlid treatment (the largest predator with the largest gape size) for most behavioral responses, this was only significant for shoaling behavior and surface water use for guppies from populations sympatric with pike cichlids. Whether this was an evolved or a learned response is unknown.

Guppies typically respond to predatory fish by increasing their shoaling behavior and moving to the surface water (Magurran 2005, Botham et al. 2006). They also inspect the predator and reduce their activity, a response commonly displayed under a wide range of different predatory threats (Magurran 2005). The extent of the predatory threat posed by freshwater prawns in the wild is uncertain, and while it has been shown that guppies inspect these predators (Magurran and Seghers 1990b), there has been little information on any other behavioral responses by guppies toward them. While we found that predatory fish elicited the usual antipredator responses, when confronted with a freshwater prawn,

guppies did not spend more time in the surface water or reduce their foraging activity compared to the control. We suggest this may be due to differences in hunting strategies between different predator taxa since guppies showed recognition of a predatory threat in all predator treatments (elevated shoaling behavior and orientation toward the predator container compared to the control). Freshwater prawns are nocturnal, have poor vision, and hunt primarily by olfaction (Magurran and Seghers 1990b). Given that foraging guppies are unlikely to be detected more easily than stationary guppies by olfactory predators (compared to visual predatory fish that preferentially attack guppies adopting the typical head-down foraging position; Krause and Godin 1996), they may not benefit by reducing their foraging activities. We might also expect that guppies should shoal less when faced with freshwater prawns, as a shoal may attract a disproportionate number of attacks from an olfactory predator (Krause and Ruxton 2002). However, we did not find this. In addition, specific antipredator responses need not necessarily be universally effective and can even make prey species more susceptible to other predators (e.g., Soluk and Collins 1988). Because of this, when prey are faced with predators with which they have no experience, they may perform more general antipredator behaviors. Another factor worth considering is the degree to which our experimental design elevated the baseline levels of each behavioral response. Novel objects such as the predator container, and an unnatural environment, often invoke some level of antipredator behavior (Brown et al. 2004). If freshwater prawns are only weak guppy predators, they may not elicit strong enough responses to show differences from the control treatment levels.

In summary, we found that predation risk varies greatly between different prey populations (both in terms of the type and number of predators present), and that this variation is stable over time. Using these measurements of variability in predation risk, we found risk-sensitive behavior is strongly affected by an interaction between the type of predators a prey is faced with and the type of predators with which it is sympatric with. While prey can respond to a whole range of predators with which they have had no prior experience, they may be unable to respond appropriately to a novel predator species that is more dangerous than those they have interacted with in the past. This has important implications for the range expansion of prey species, since, when colonizing new habitats, they are likely to encounter new predators. Similarly, predator species may expand their ranges making native prey species vulnerable as they may be unable to respond appropriately. Historically, many reintroduction programs and attempts to control alien species have encountered problems and have resulted in limited success. Reintroductions involve the release of either captive-bred animals or animals from a population previously occupying a different habitat region into an environ-

ment in which they have become extinct or close to extinct. In these cases, they are likely to confront predators with which they have had no prior experience, and thus, are unable to respond effectively. In addition, the number and type of predators present in many environments varies temporally due to the seasonal and cyclical variation in habitat suitability (Bronmark and Miner 1992, Craig 1996). Recent changes in the climate make it likely that many potential predator species may expand or shift their ranges into habitats/regions they were previously not found in (Parmesan and Yohe 2003). We suggest that the success of reintroductions may be enhanced if prey are given prior experience with the full range of predators that they are likely to encounter both at the time of release and in the future, accounting for range expansions of predator species as a result of climate change and seasonal variation in habitat suitability. Similarly, the response of native prey species to exotic predators should be considered for the control of invasive alien species, given that many invasive alien species are predators themselves, and thus, the prey that they will encounter are unlikely to have developed the appropriate antipredator responses (e.g., Goudswaard et al. 2008).

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APPENDIX

A table of comparisons of predator treatments where no significant interaction terms were found between a treatment and population predation risk (*Ecological Archives* E089-184-A1).